

D790

Human Centered AI

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital deploys an AI that recommends stopping a patient's medication. The vendor's model won an industry award last year. A wrong recommendation could seriously harm a patient. What level of human oversight fits?

- A. Trust the model because it won an award.
- B. Human in the loop: a clinician reviews and approves each recommendation.
- C. Human on the loop: staff watch weekly dashboards of outcomes.
- D. Fully automated, with changes applied directly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An email spam filter handles millions of messages a day. Errors are low-stakes, and users can easily move a message back. What oversight fits?

- A. Automated decisions with humans on the loop monitoring error rates and user corrections.
- B. A person approves every message before it is filtered.
- C. No monitoring at all, ever.
- D. Human in the loop, since oversight and approval mean the same thing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Loan officers using an AI risk score approve 97% of its recommendations, even when the file shows obvious errors the AI missed. The bank recently repainted its branches. Which change best reduces this problem?

- A. Drop the officers' review since they agree anyway.
- B. Train officers to trust the score more.
- C. Update the branch décor to feel more modern.
- D. Have officers record their own assessment before seeing the score, and show the score's key reasons.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which observation most strongly suggests automation bias?

- A.** Users reject AI suggestions that are usually correct.
- B.** The AI has high accuracy on test data.
- C.** Users accept AI outputs that conflict with clear evidence in front of them.
- D.** Users sometimes override the AI after checking the facts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An AI system denies a small-business loan, and the applicant asks why. The bank's data team offers to send the model's 400-feature weight table. The applicant runs a neighborhood bakery. What explanation fits best?

- A.** The main reasons in plain language and what could change the outcome.
- B.** The full 400-feature weight table.
- C.** A summary of trends in the bakery industry.
- D.** No explanation, because models are always proprietary.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An AI résumé screener advanced 60 of 200 applicants from Group A and 18 of 90 from Group B. The tool was built by a well-funded startup. Under the four-fifths rule, is there a sign of adverse impact?

- A.** Yes; $18 \div 60 = 0.30$.
- B.** No; $30\% \div 20\% = 1.5$, above 0.8.
- C.** No; a well-funded vendor's tool can be trusted.
- D.** Yes; $20\% \div 30\% \approx 0.67$, below 0.8.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Group A's selection rate is 50%. What is the lowest Group B selection rate that avoids a four-fifths flag?

- A.** 80%
- B.** 40%
- C.** 30%
- D.** 62.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A lender removed race from its model's inputs, but approval rates still differ sharply by race. The model uses ZIP code and shopping history among its 30 features. What is the most likely explanation?

- A. The model must be reading race from a hidden field.
- B. Features like ZIP code act as proxies for race.
- C. The model has too few features.
- D. It cannot be bias, since removing race guarantees fairness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team wants to check whether its model treats age groups fairly, but the age column was deleted from its data. What should it do?

- A. Nothing; deleting age makes the model fair.
- B. Never collect protected attributes for any purpose.
- C. Keep age data under strict governance for auditing outcomes, even if it is not a model input.
- D. Add age back as a model input.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A routing AI gives delivery drivers one route with no sign of how confident it is. Drivers either follow it blindly or ignore it completely. It launched in spring with a new logo. Which change best helps drivers judge when to rely on it?

- A. Show a confidence level and a short reason with each route.
- B. Relaunch with new branding.
- C. Offer three routes with no confidence shown.
- D. Always display high confidence to build trust.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A gig platform’s AI automatically deactivates driver accounts for low ratings, and drivers cannot reach a person. The platform operates in 12 cities. What is most needed?

- A. Operating in fewer cities.
- B. An automated email explaining the deactivation.
- C. A lower rating threshold for deactivation.
- D. An appeal path with human review and a way to correct wrong data.

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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12.

A city is designing an AI to rank housing-assistance applications. The project team includes data scientists and IT staff, and the budget is \$2 million. Whom should it involve, and when?

- A. Increase the budget before involving anyone new.
- B. Only technical staff, because it is a technical system.
- C. Applicants, caseworkers, and advocates, from the start and throughout design.
- D. Residents through a complaint form after launch.

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A hospital deploys an AI that recommends stopping a patient's medication. ~~The vendor's model won an industry award last year.~~ A wrong recommendation could seriously harm a patient. What level of human oversight fits?

Target word: oversight level **Decides:** Serious, possibly irreversible harm.

A	Camouflage	An award does not remove the need for case-level oversight.
B	Correct	High-stakes decisions need a person to decide each case.
C	Adjacent category	Monitoring catches trends, not individual harmful decisions.
D	Inverted operation	Removes the human where the stakes are highest.

2. Answer A Concept 1 • transfer

An email spam filter handles millions of messages a day. Errors are low-stakes, and users can easily move a message back. What oversight fits?

Target word: oversight for low-stakes decisions **Decides:** Low stakes and easy to reverse.

A	Correct	Reversible, low-stakes decisions do not need case-by-case approval.
B	Inverted operation	Case-by-case approval fits high stakes, not this.
C	Absolute language	Even low-stakes systems need monitoring for drift and errors.
D	Confusion set	Monitoring and approving each decision are different levels.

3. Answer D Concept 2

Loan officers using an AI risk score approve 97% of its recommendations, even when the file shows obvious errors the AI missed. ~~The bank recently repainted its branches.~~ Which change best reduces this problem?

Target word: reduce automation bias **Decides:** Reviewers defer to the AI against clear evidence.

A	Dropped element	Removes the check that should catch AI errors.
B	Inverted operation	Deepens the over-reliance.
C	Camouflage	Décor has nothing to do with review behavior.
D	Correct	Forces independent judgment and exposes weak reasoning.

4. Answer C Concept 2 • transfer

Which observation most strongly suggests automation bias?

Target word: recognize the bias **Decides:** Accepting AI output against clear contrary evidence.

A	Inverted operation	That is the opposite problem, distrust of algorithms.
B	Adjacent category	Describes the model, not human behavior.
C	Correct	That is over-reliance on automation.
D	Confusion set	That is healthy, calibrated oversight.

5. Answer A Concept 3

An AI system denies a small-business loan, and the applicant asks why. The bank's data team offers to send the model's 400-feature weight table. ~~The applicant runs a neighborhood bakery.~~ What explanation fits best?

Target word: useful explanation **Decides:** A non-expert needs reasons and recourse.

A	Correct	Meaningful explanations are understandable and actionable.
B	Confusion set	Technical transparency without meaning to the reader.
C	Camouflage	Industry context does not explain this decision.
D	Absolute language	Affected people should get reasons for consequential decisions.

6. Answer D Concept 4

An AI résumé screener advanced 60 of 200 applicants from Group A and 18 of 90 from Group B. ~~The tool was built by a well-funded startup.~~ Under the four-fifths rule, is there a sign of adverse impact?

Target word: four-fifths check **Decides:** Rates 30% and 20%.

A	Wrong stem number	Compares raw counts instead of rates.
B	Inverted operation	Divides the higher rate by the lower one.
C	Camouflage	Funding does not settle fairness.
D	Correct	Group B's rate is about two-thirds of Group A's.

7. Answer B Concept 4 · transfer

Group A's selection rate is 50%. What is the lowest Group B selection rate that avoids a four-fifths flag?

Target word: solve for the threshold rate **Decides:** $0.8 \times 50\%$.

A	Wrong stem number	The ratio, not the rate.
B	Correct	$0.8 \times 50\% = 40\%$.
C	Confusion set	Subtracts 20 points instead of taking 80%.
D	Inverted operation	Divides 50% by 0.8.

8. Answer B Concept 5

A lender removed race from its model's inputs, but approval rates still differ sharply by race. The model uses ZIP code and shopping history ~~among its 30 features~~. What is the most likely explanation?

Target word: why bias persists **Decides:** Other features correlate with race.

A	Adjacent category	No hidden field is needed when proxies exist.
B	Correct	Correlated features carry the same information.
C	Camouflage	Feature count is not the cause.
D	Absolute language	Fairness through unawareness fails with proxies.

9. Answer C Concept 5 · transfer

A team wants to check whether its model treats age groups fairly, but the age column was deleted from its data. What should it do?

Target word: enable a fairness audit **Decides:** You cannot measure gaps without group data.

A	Dropped element	Ignores proxies and leaves bias unmeasured.
B	Absolute language	Blocks the measurement fairness requires.
C	Correct	Outcome checks by group need the group attribute.
D	Adjacent category	Auditing needs age, not using it to decide.

10. Answer A Concept 6

A routing AI gives delivery drivers one route with no sign of how confident it is. Drivers either follow it blindly or ignore it completely. ~~It launched in spring with a new logo.~~ Which change best helps drivers judge when to rely on it?

Target word: support calibrated trust **Decides:** No confidence information is shown.

A	Correct	Lets drivers rely on it when strong and question it when weak.
B	Camouflage	Branding does not inform reliance.
C	Dropped element	More choices still lack the missing signal.
D	Absolute language	False certainty makes trust less calibrated.

11. Answer D Concept 7

A gig platform's AI automatically deactivates driver accounts for low ratings, and drivers cannot reach a person. ~~The platform operates in 12 cities.~~ What is most needed?

Target word: missing safeguard **Decides:** Automated adverse action with no human contact.

A	Camouflage	Scale is not the issue.
B	Dropped element	Explains but offers no way to challenge.
C	Adjacent category	Changes the rule, not the lack of recourse.
D	Correct	Contestability gives affected people recourse.

12. Answer C Concept 8

A city is designing an AI to rank housing-assistance applications. The project team includes data scientists and IT staff, ~~and the budget is \$2 million.~~ Whom should it involve, and when?

Target word: who and when in the design process **Decides:** Affected people are missing from the team.

A	Camouflage	Budget does not decide stakeholder involvement.
B	Dropped element	Leaves out the people it affects.
C	Correct	Human-centered design involves those affected early and iteratively.
D	Inverted operation	Feedback after launch is too late to shape the design.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Matching oversight to stakes	High-stakes, hard-to-reverse decisions need a human in the loop approving each one; low-stakes, reversible ones can run with a human on the loop monitoring.	Using one level of oversight for every AI system.	How severe and reversible a wrong decision is.
2	Automation bias	People tend to over-accept AI output; design must prompt independent judgment and make disagreement easy.	Assuming a human reviewer automatically catches AI errors.	Reviewers agreeing with the AI even against clear evidence.
3	Meaningful explanation	An explanation should give the affected person the main reasons in plain terms and what could change the outcome.	Equating transparency with dumping model internals.	A non-expert asking why a decision went against them.
4	Disparate impact (four-fifths rule)	If one group's selection rate is below 80% of the highest group's rate, that signals possible adverse impact.	Comparing raw counts or dividing the wrong way.	Selections and applicants for two groups.
5	Proxy variables	Removing a protected attribute does not remove bias when other features stand in for it; fairness must be checked on outcomes.	Fairness through unawareness.	A protected attribute removed but gaps remain.
6	Calibrated trust	Showing confidence and reasons helps users trust AI when it is reliable and doubt it when it is not.	Projecting certainty to increase adoption.	Users either blindly follow or wholly ignore the AI.
7	Contestability and recourse	People affected by automated decisions need a way to appeal to a human and correct the data used.	Offering an automated notice as if it were recourse.	An automated adverse action with no human contact.
8	Human-centered design process	Involve the people affected, not just builders and buyers, from the start and iterate with them.	Collecting feedback only after launch.	A project team made only of technical staff.

Course scope (public description): This course provides students with the opportunity to learn about how AI systems are designed to align with human behavior and values. Students will characterize human centered design as it relates to AI, while analyzing the ethical implications of AI solutions on both individuals and society. Students gain insights into the impact of AI on humans and human values. Finally, students will understand and explain how to incorporate mechanisms for human oversight and control in AI systems. This learning will focus on the deep analysis of software engineering as it relates to complex understanding of human/AI interactions.